

TechBiz Security

Cyber Security Internship Program

WEEK 5 – TASK 13 (SOC)

PHISHING EMAIL ANALYSIS

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3. Internship Task Overview

This task focused on phishing email investigation from a SOC Level 1 (L1) analyst perspective. The exercise involved reviewing sample phishing emails, examining email headers and authentication results, analyzing sender and domain information, safely investigating suspicious URLs using reputation-based tools, and documenting the findings in a structured manner consistent with SOC reporting practices.

4. Task Objective

The objective of this task was to analyze email metadata, authentication results, sender information, and URLs, and to identify observable phishing indicators, while following safe investigation practices. This included avoiding direct interaction with suspicious links or attachments and relying instead on static analysis and reputation-checking tools such as VirusTotal.

5. Tools and Resources Used

- TryHackMe – Phishing Analysis Fundamentals
- Mozilla Thunderbird
- VirusTotal
- Email sample files provided in the TryHackMe lab

6. Lab Environment

The practical exercise was carried out within the TryHackMe “Phishing Analysis Fundamentals” lab environment. The lab provided a set of sample email files located in an Email Samples directory, consisting of email1.eml, email2.txt, and email3.eml. These files served as the source evidence for the header and content analysis documented in this report.

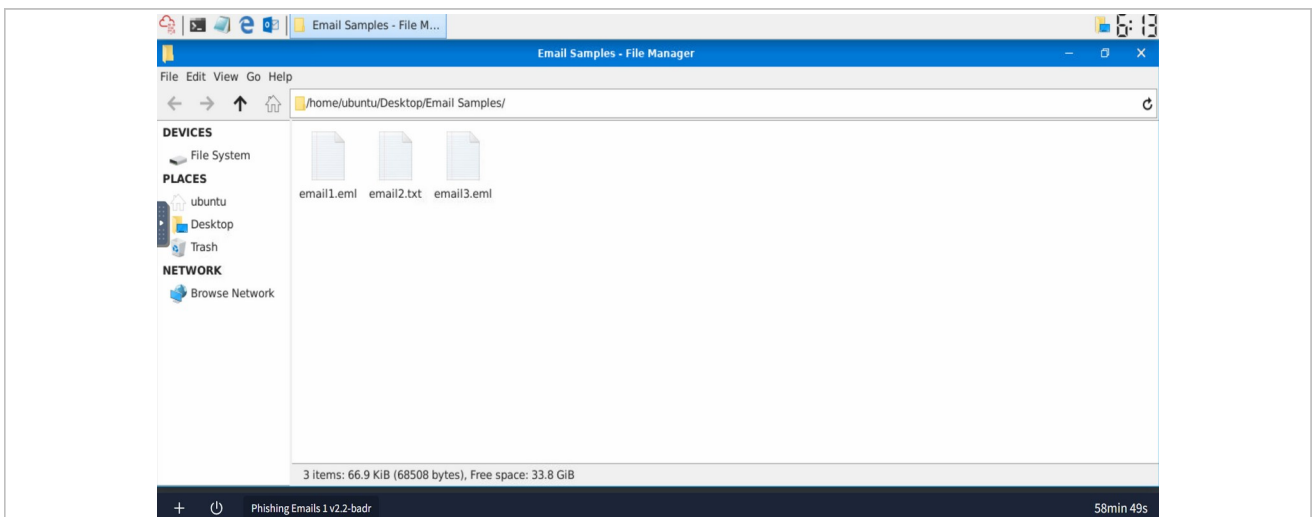


Figure 4: Email Samples directory showing email1.eml, email2.txt, and email3.eml.

7. Phishing Email Analysis

7.1 Email Sample 1 – ant.anki-tech.com (Reference)

Figure 1 shows the first sample email, used as a reference point for comparison with the header, authentication, and URL reputation results discussed in later sections of this report.

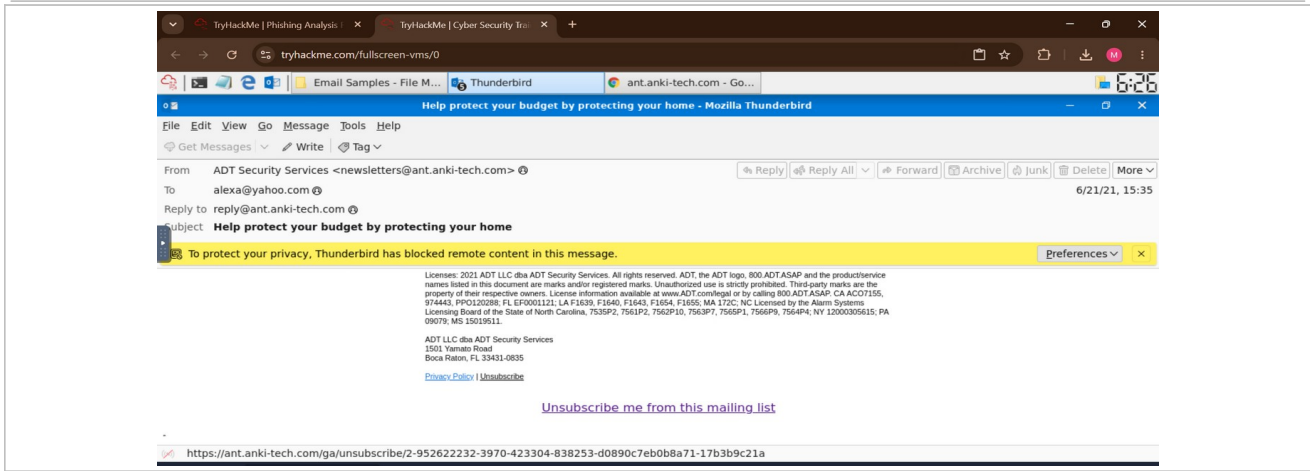


Figure 1: Phishing email sample – ADT Security Services newsletter (ant.anki-tech.com).

The email displays a sender of “ADT Security Services <newsletters@ant.anki-tech.com>”, addressed to alexa@yahoo.com, with the subject “Help protect your budget by protecting your home” and a Reply-To address of reply@ant.anki-tech.com. The message contains an unsubscribe link and references ADT Security Services in its body content.

7.2 Email Sample 2 – Suspicious Phishing Email (teckbe.com)

The suspicious email shown in Figure 6 was reviewed as part of this task. The email was opened in Mozilla Thunderbird for static inspection only; no links or attachments were clicked or executed.

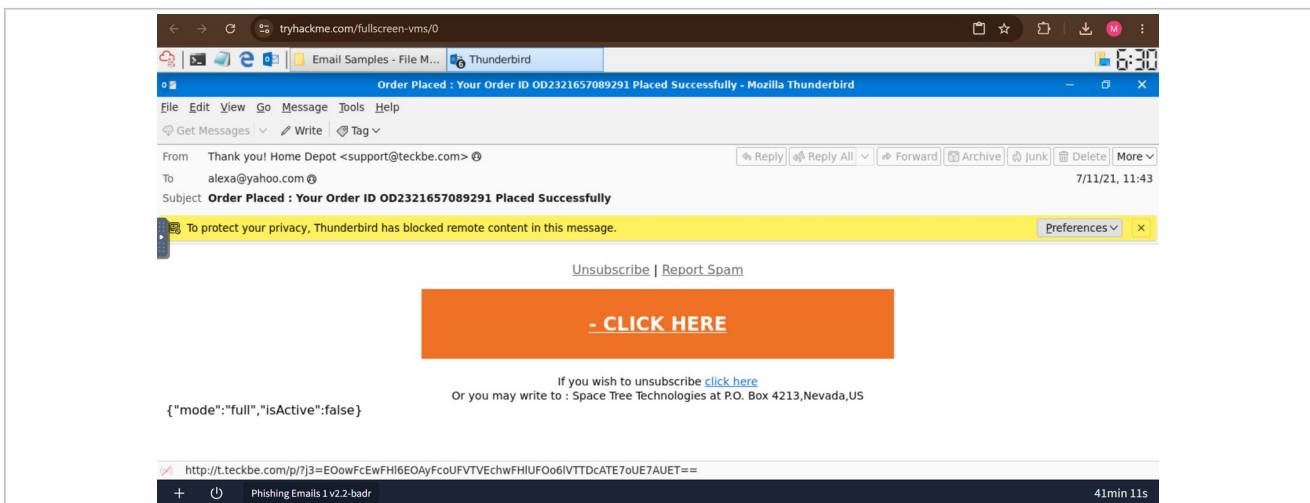


Figure 6: Suspicious phishing email displayed in Mozilla Thunderbird.

Field	Observation	Security Relevance
Sender (display name)	Thank you! Home Depot	Display name references a well-known retail brand, unrelated to the visible sending domain.
Sender email address	support@teckbe.com	Domain does not match the Home Depot brand referenced in the display name.
Recipient	alexa@yahoo.com	Not observed to be relevant beyond confirming the message destination.
Subject	Order Placed : Your Order ID OD2321657089291 Placed Successfully	Order-related subject line designed to attract attention regarding an unexpected purchase.
Visible links	"- CLICK HERE -" button link and a separate "click here" unsubscribe link	Prominent call-to-action link embedded in the email body.
Domain (link)	t.teckbe.com (from the "- CLICK HERE -" link,	Link domain does not align with the Home

Field	Observation	Security Relevance
	visible in the status bar)	Depot brand referenced in the sender name.
Brand/reference	“Home Depot” in sender display name; “Space Tree Technologies” at P.O. Box 4213, Nevada, US in the footer	Mismatch between the brand referenced in the display name and the organization named in the footer.
Other suspicious characteristics	“Unsubscribe” and “Report Spam” options, blocked remote content warning from Thunderbird	Bulk-mail characteristics combined with sender/domain inconsistency.

8. Email Header Analysis

Figure 5 shows the raw header information extracted from email1.eml. The relevant header fields are summarized below.

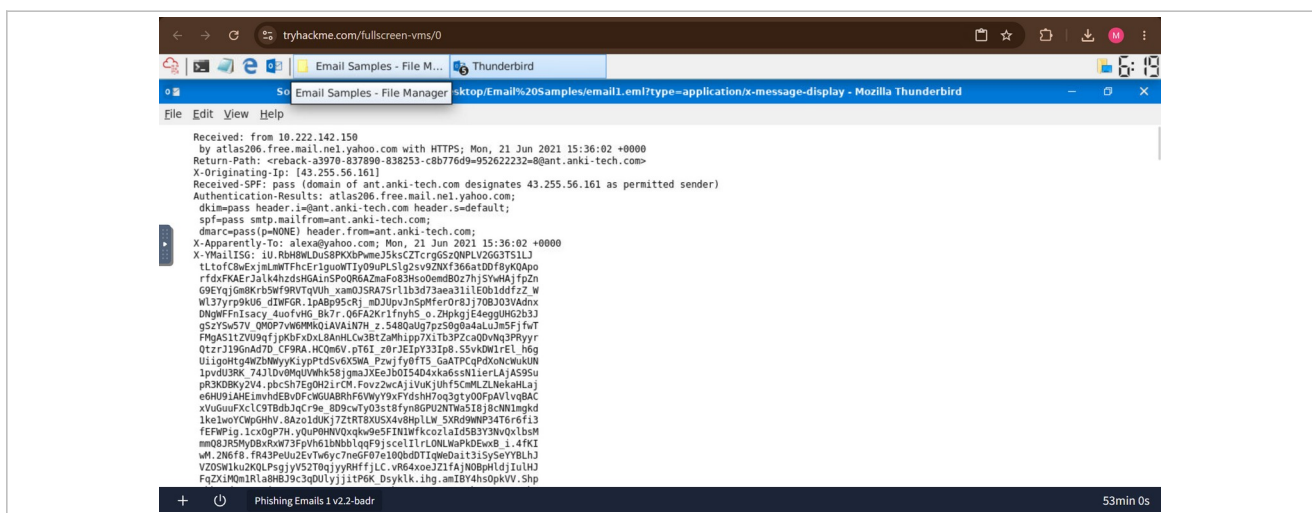


Figure 5: Raw email header analysis for email1.eml.

Header / Authentication Field	Observed Value	Security Interpretation
Received	from 10.222.142.150 by atlas206.free.mail.ne1.yahoo.com	Indicates the internal mail-relay hop recorded by the receiving mail system.
Return-Path	<reback-a3970-837890-838253-c8b776d9=952622232=8@ant.anki-tech.com>	Address to which bounce messages are returned; associated with the ant.anki-tech.com domain.
X-Originating-IP	[43.255.56.161]	IP address recorded by the mail system as the originating source of the message.
SPF	Received-SPF: pass (domain of ant.anki-tech.com designates 43.255.56.161 as permitted sender)	Indicates the sending IP was authorized to send mail for the ant.anki-tech.com domain.
DKIM	dkim=pass header.i=@ant.anki-tech.com header.s=default	Indicates the message body and headers were signed and the signature validated against the stated domain.
DMARC	dmarc=pass (p=NONE) header.from=ant.anki-tech.com	Indicates the message passed DMARC alignment checks, with a policy of “none” applied.

The header values above are taken directly from the screenshot in Figure 5. SPF, DKIM, and DMARC all report a “pass” result for the ant.anki-tech.com domain. This confirms that the sending infrastructure was authorized to send mail on behalf of that domain and that the message was not altered in transit for the fields covered by DKIM signing. It does not, however, confirm that the email content itself is safe, trustworthy, or free of phishing intent – authentication and content trustworthiness are separate concerns.

9. Sender and Domain Analysis

A sender display name and the underlying email domain should always be examined separately, since a display name can be set to any arbitrary text regardless of the actual sending domain. For the email shown in Figure 6, the sender field displays:

Thank you! Home Depot <support@teckbe.com>

The display name references the Home Depot brand, while the actual sending domain is teckbe.com, which has no visible relationship to Home Depot. This creates an observable impersonation and domain-mismatch concern: a recipient scanning only the display name could reasonably associate the message with the Home Depot brand, while the underlying domain does not support that association. Based on the evidence available, it is not possible to confirm domain ownership or intent; this observation is presented strictly as an observable indicator rather than a confirmed finding of malicious activity.

10. SPF, DKIM and DMARC Analysis

The header evidence in Figure 5 shows the following authentication results for the ant.anki-tech.com email (Figure 1):

- SPF: pass
- DKIM: pass
- DMARC: pass

A “pass” result for SPF indicates the sending IP address was authorized by the domain's SPF record. A “pass” result for DKIM indicates the message carried a valid cryptographic signature associated with the stated domain. A “pass” result for DMARC indicates that the message met the domain's alignment requirements between the visible “From” domain and the authenticated sending domain. Together, these results confirm that the sending infrastructure was properly authorized and aligned for the ant.anki-tech.com domain. This authentication outcome does not, by itself, establish that the email content is legitimate, safe, or free of phishing characteristics; a domain can pass all three checks while still being used to send unwanted or deceptive mail.

11. Suspicious URL Investigation

Two separate URLs were investigated using VirusTotal reputation lookups. In both cases, the URL was submitted for reputation analysis and was not directly visited or executed.

11.1 URL Investigation A – ant.anki-tech.com

- Domain: ant.anki-tech.com
- VirusTotal displayed result: 0/89 detections

11.2 URL Investigation B – t.teckbe.com

- URL: http://t.teckbe.com/p/?j3=... (truncated in the source screenshot)
- Domain: t.teckbe.com
- IP address: 83.167.244.202
- Content type: text/html
- VirusTotal displayed result: 0/95 detections
- Last analysis date: 7 months ago

These two findings are reported separately and were not merged, as they relate to different domains, different VirusTotal reports, and different email samples.

12. VirusTotal Analysis

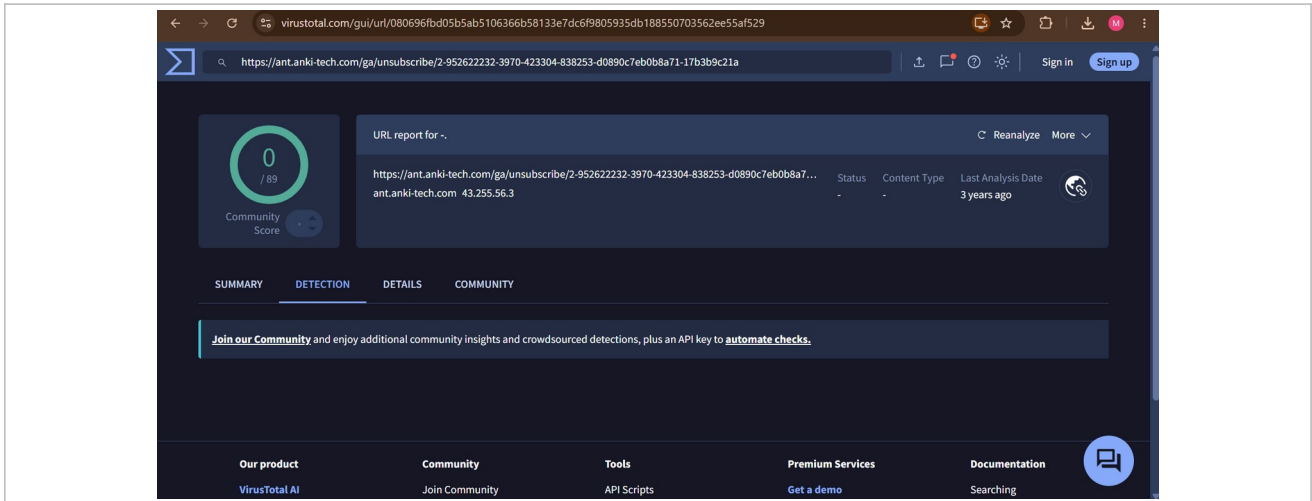


Figure 2: VirusTotal URL analysis for the ant.anki-tech.com unsubscribe link.

For the ant.anki-tech.com URL, VirusTotal displayed a result of 0/89, meaning that none of the 89 security vendors queried at the time of analysis flagged the submitted URL as malicious. The screenshot also shows the associated IP address, 43.255.56.3, and indicates that the URL had a prior analysis on record.

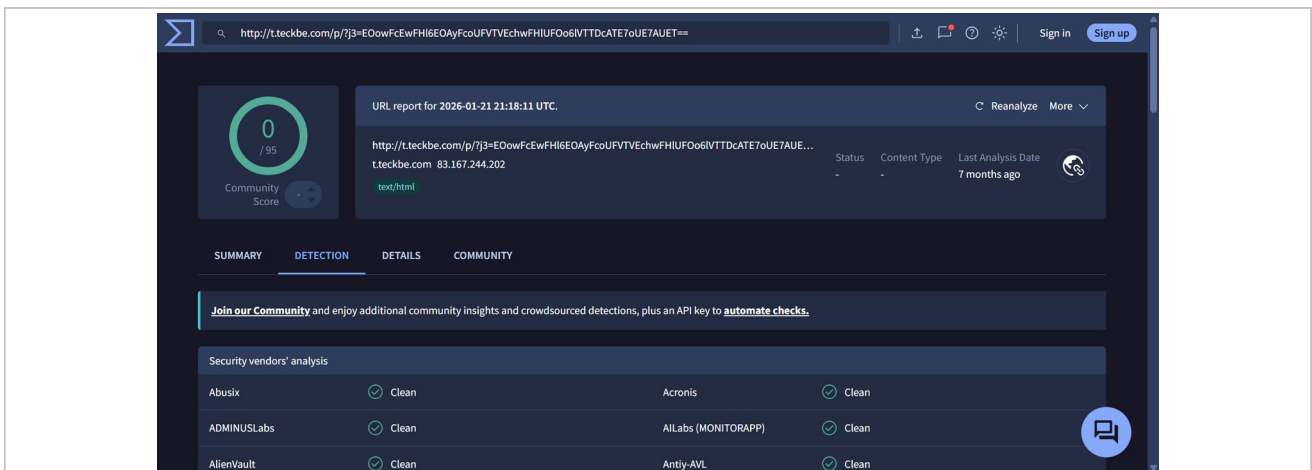


Figure 7: VirusTotal URL analysis for the t.teckbe.com link.

For the t.teckbe.com URL, VirusTotal displayed a result of 0/95, meaning that none of the 95 security vendors queried at the time of analysis flagged the submitted URL as malicious. The report additionally shows the domain (t.teckbe.com), IP address (83.167.244.202), content type (text/html), and a last analysis date of 7 months ago, with the visible vendor results showing multiple “Clean” entries.

In both cases, a result of 0 detections means the submitted URL was not flagged by the displayed security vendors at the time of that analysis. This is not, by itself, sufficient to classify the associated email as legitimate. Reputation databases depend on prior submissions and vendor coverage, and a newly created or low-volume malicious link may show no detections despite being used for phishing. VirusTotal results should therefore be treated as one input among several, not as a standalone verdict.

13. Phishing Indicators / Red Flags

Indicator	Evidence	Risk/Significance
Brand impersonation concern	Sender display name “Thank you! Home Depot” used with an unrelated domain	May mislead a recipient into associating the message with a trusted brand.
Sender/domain mismatch	support@teckbe.com used under the Home Depot display name	Indicates the visible brand and the actual sending domain do not correspond.
Suspicious or unfamiliar domain	teckbe.com / t.teckbe.com	Domain unrelated to the referenced brand and not independently verified as trustworthy.
Prominent “CLICK HERE” link	Large orange “- CLICK HERE -” button in the email body	Encourages the recipient to click without independently verifying the order or sender.
Unsubscribe links	“Unsubscribe” and “click here” links present in the email	Common in bulk/marketing mail and can also be used in phishing campaigns to harvest clicks.
Order-related subject line	“Order Placed : Your Order ID OD2321657089291 Placed Successfully”	Designed to attract attention from a recipient who has not placed such an order.
Potentially misleading sender display name	“Thank you! Home Depot” display text	Display names can be freely set and do not verify the true sending organization.
URL requiring reputation analysis	http://t.teckbe.com/p/?j3=...	Link destination required a reputation lookup before any further evaluation could be made.

14. Email Security Assessment

The email shown in Figure 1 (ant.anki-tech.com, ADT Security Services newsletter) passed SPF, DKIM, and DMARC authentication checks as shown in Figure 5, and the associated unsubscribe URL returned 0/89 detections on VirusTotal. Based strictly on the available evidence, this email does not display the same sender/domain mismatch or brand-impersonation characteristics observed in the second sample.

The email shown in Figure 6 (support@teckbe.com, displayed as “Thank you! Home Depot”) presents an observable mismatch between the referenced brand and the sending domain, combined with a prominent call-to-action link and an unexpected order notification. The associated link (t.teckbe.com) returned 0/95 detections on VirusTotal; this result does not confirm the link is safe, and should be considered alongside the sender/domain and content-based indicators identified above.

15. Final Verdict

For the email in Figure 1 (ant.anki-tech.com): based on the available evidence – passing SPF/DKIM/DMARC results and a clean VirusTotal result for the associated link – no clear phishing indicators were identified. This assessment is limited to the evidence reviewed and does not constitute a guarantee of legitimacy.

For the suspicious email in Figure 6 (teckbe.com / Home Depot): based on the observable indicators – apparent brand impersonation, sender/domain inconsistency, a suspicious link structure, a domain requiring further investigation, and email context indicating an unexpected order notification – the email should be treated as suspicious and handled as a potential phishing attempt pending further verification. The VirusTotal result of 0/95 detections does not confirm the URL as malicious and is not used as the basis for this verdict; the verdict rests on the sender, domain, and content indicators described above.

16. SOC Analyst Recommendations

- Do not click suspicious links.
- Do not open unknown attachments.
- Preserve the original email and headers.
- Defang suspicious URLs when documenting them.
- Verify the sender through an independent trusted channel.
- Check domain reputation and authentication results.
- Escalate suspicious emails according to organizational SOC procedures.
- Block/quarantine confirmed malicious indicators where appropriate.
- Educate users about sender-name impersonation and suspicious links.

17. TryHackMe Practical Training

The mandatory TryHackMe training room was completed as part of this task.

- Training Room: Phishing Analysis Fundamentals
- Completion: 100%

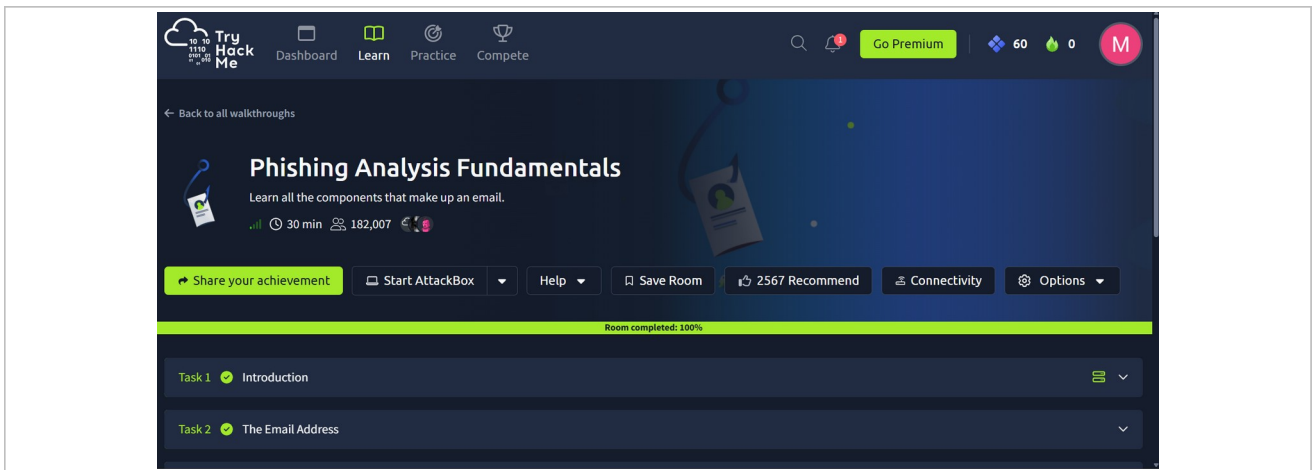


Figure 3: TryHackMe “Phishing Analysis Fundamentals” room completed at 100%.

18. Evidence and Screenshots

Figure	Screenshot Description	Purpose
Figure 1	Phishing Email Sample (ADT Security Services, ant.anki-tech.com)	Shows the sender, recipient, subject, and unsubscribe link of the first sample email.
Figure 2	VirusTotal URL Analysis for ant.anki-tech.com	Provides reputation analysis results (0/89 detections) for the unsubscribe link.
Figure 3	TryHackMe Room Completion	Evidences 100% completion of the Phishing Analysis Fundamentals room.
Figure 4	Email Samples Directory	Shows the sample email files (email1.eml, email2.txt, email3.eml) provided in the lab.
Figure 5	Email Header Analysis	Shows raw header data for email1.eml, including SPF, DKIM, and DMARC results.
Figure 6	Suspicious Phishing Email (Home Depot / teckbe.com)	Shows the sender, subject, and suspicious link content of the second sample email.
Figure 7	Suspicious URL Analysis for t.teckbe.com	Provides reputation analysis results (0/95

Figure	Screenshot Description	Purpose
		detections) for the suspicious link.

19. Key Findings

- The email in Figure 1 (ant.anki-tech.com) passed SPF, DKIM, and DMARC authentication checks.
- The unsubscribe link associated with ant.anki-tech.com returned 0/89 detections on VirusTotal.
- The email in Figure 6 displays a sender/domain mismatch: “Thank you! Home Depot” display name paired with the teckbe.com domain.
- The teckbe.com-linked URL (t.teckbe.com) returned 0/95 detections on VirusTotal, which does not by itself confirm the link is safe.
- The Figure 6 email contains a prominent “CLICK HERE” link and an unexpected order notification, both of which are observable phishing-style characteristics.
- The TryHackMe “Phishing Analysis Fundamentals” room was completed at 100%, as shown in Figure 3.
- All URL investigation in this task was performed through VirusTotal reputation lookups, without directly visiting the suspicious links.

20. Conclusion

This exercise demonstrated a practical SOC L1 phishing-analysis workflow involving email inspection, header analysis, authentication verification, sender/domain analysis, URL reputation checking, phishing indicator identification, and an evidence-based verdict. All suspicious content was handled safely, using static analysis and reputation-based tools rather than direct interaction with links or attachments.

The task reinforced the importance of examining sender display names and domains separately, understanding the scope and limitations of SPF, DKIM, and DMARC authentication results, and recognizing that a clean VirusTotal result does not, on its own, establish that an email or URL is safe. Completion of the TryHackMe Phishing Analysis Fundamentals room further supported the theoretical understanding applied during the practical investigation.